

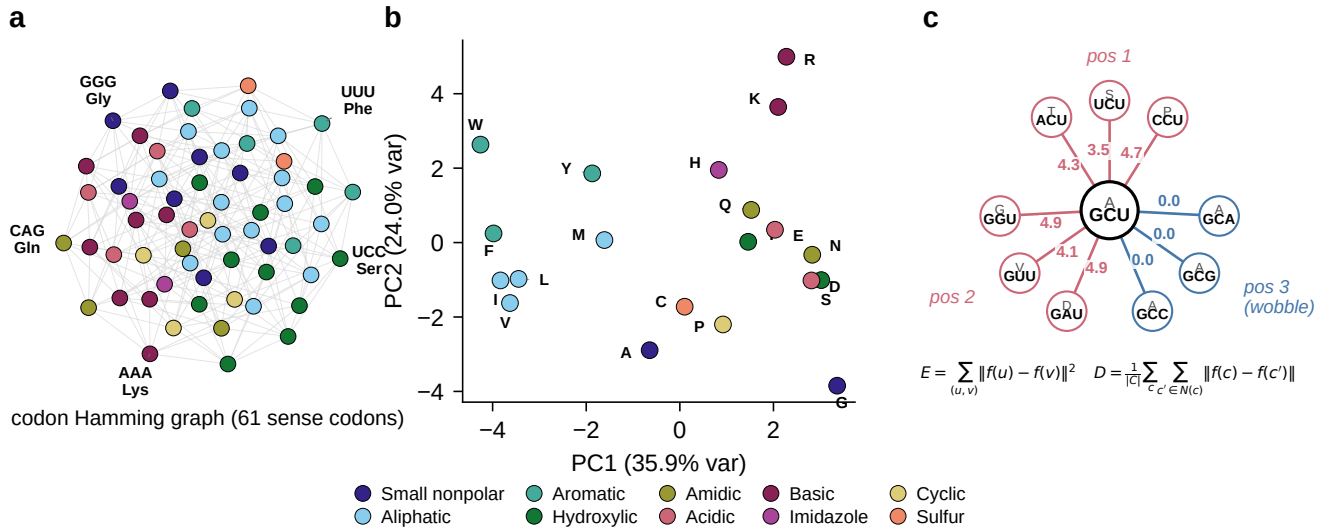


Figure 1. Framework. (a) Codon Hamming graph for $n = 3$: all 61 sense codons, with edges connecting codons that differ by a single nucleotide substitution (264 edges; average degree 8.6). Nodes are colored by amino acid chemical class (10-class RAA10 grouping, Murphy 2000); five peripheral anchor codons are labeled for orientation. (b) 20 amino acids in PC1-PC2 space of the 22-property descriptor set (after correlation-pruning and z-scoring; PC1 = 35.9 % of variance, PC2 = 24.0 %; 97.1 % total in the retained eight PCs used downstream). Hydrophobic residues (Phe, Trp, Tyr, Ile, Leu, Val, Met) lie at negative PC1; charged residues (Lys, Arg, Asp, Glu) occupy positive PC1. Class color scheme matches panel (a). (c) Hamming-1 neighborhood of the Ala codon GCU with edge-wise property distances $\|\Delta f\|$ computed in the 8-PC space: all three position-3 neighbors (GCA, GCC, GCG) are synonymous with $\|\Delta f\| = 0$ (blue edges); the six position-1 and position-2 neighbors are non-synonymous with $\|\Delta f\|$ ranging from 3.5 (UCU → Ser) to 4.9 (GGU → Gly, GAU → Asp) (red edges). Dirichlet energy E sums squared $\|\Delta f\|$ over all graph edges; noise distortion D averages $\|\Delta f\|$ over non-synonymous neighbors per codon.

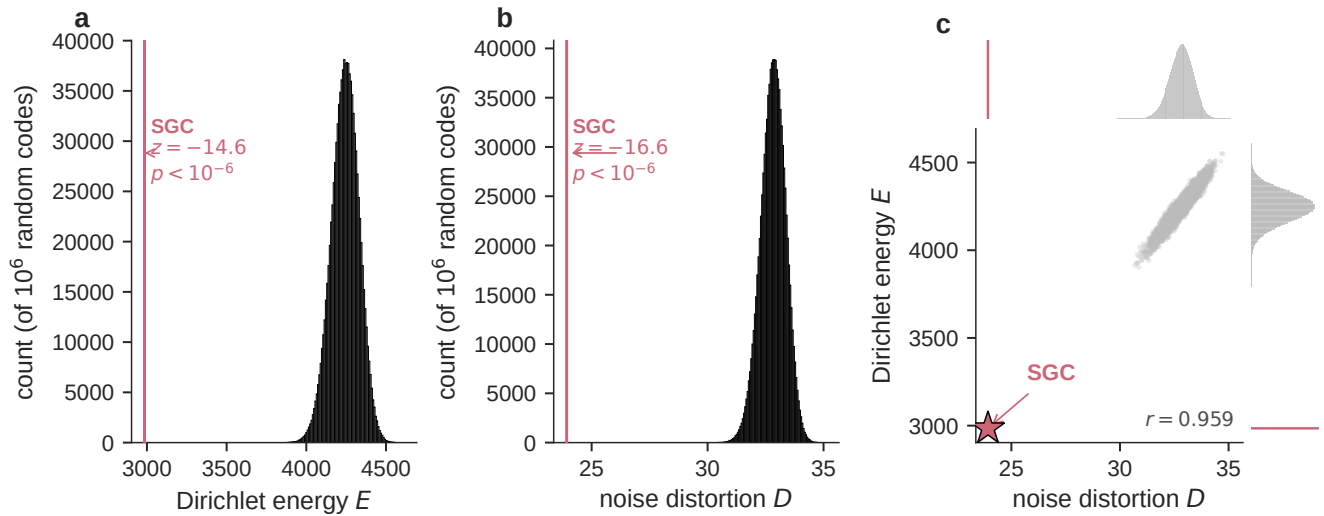


Figure 2. The Standard Genetic Code (SGC) is beyond the null under both metrics. (a) Null distribution of Dirichlet energy E across 10^6 degeneracy- preserving random codes (grey). The SGC (red line) falls at $z = -14.6$ ($p < 10^{-6}$; 0 / 10^6 random codes reach the SGC's energy). The null is generated by permuting amino acid labels across the 61 sense codons while preserving per-amino-acid codon counts, fixing stops, and pinning AUG $\rightarrow$ Met. (b) Same null-hypothesis test for noise distortion D : SGC at $z = -16.6$ ($p < 10^{-6}$; 0 / 10^6). (c) Joint scatter of (D, E) for a 10 000-point random subsample from the null distribution (grey points). The SGC (red star) lies outside and below the null cloud in both metrics simultaneously. The two metrics are strongly but imperfectly correlated across the null ($r = 0.959$), so each provides partially independent evidence; neither suggests the SGC's optimization is an artefact of the other. Marginal histograms along the top (D) and right (E) axes reproduce panels (a) and (b) for direct comparison.

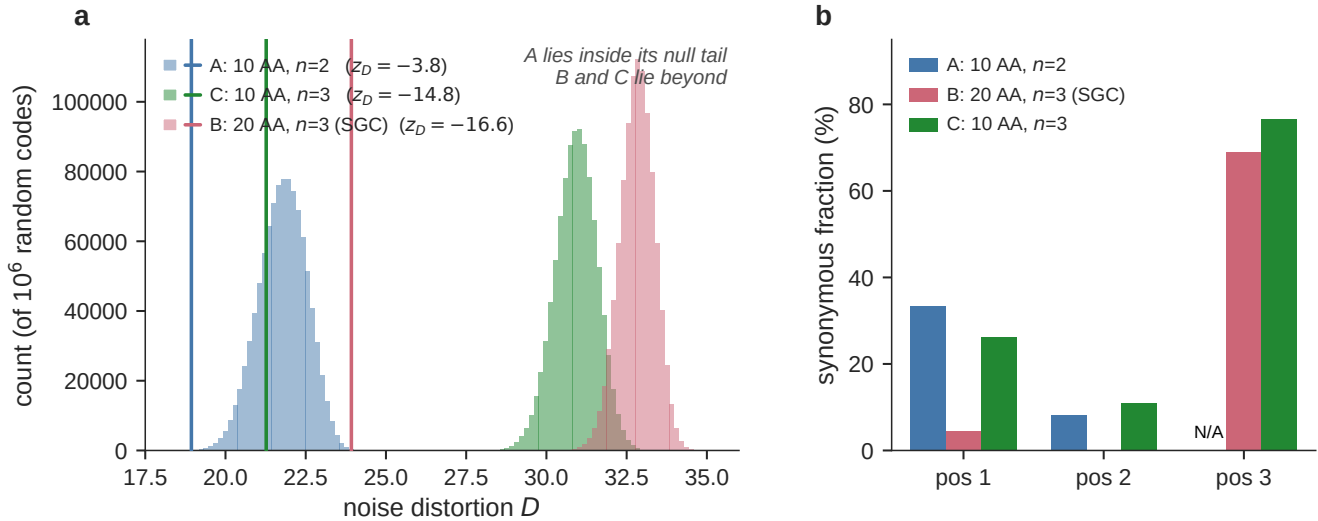
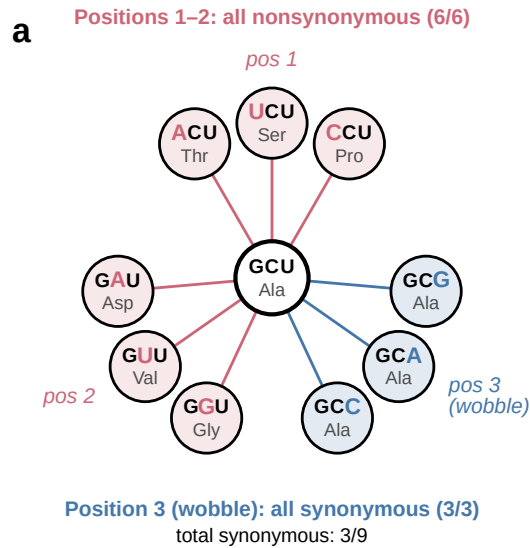


Figure 3. Triplet architecture enables deep optimization; reduced alphabet size does not. (a) Null distributions of noise distortion D across 10^6 degeneracy-preserving random codes for three conditions, on a shared x-axis: Condition A (blue; 10 amino acids, $n = 2$ doublet code obtained by majority- vote projection of the SGC onto the first two codon positions under the RAA10 reduced alphabet), Condition C (green; 10 amino acids, $n = 3$ triplet with RAA10 applied to the sense codons), and Condition B (red; 20 amino acids, $n = 3$ — the SGC itself). Vertical lines mark the observed value for the corresponding code. The doublet SGC projection (A) sits well within its null tail ($z_D = -3.8$); the triplet conditions both lie beyond their nulls ($z_D = -14.8$ for C, $z_D = -16.6$ for B). The B and C null distributions substantially overlap, indicating the architectural shift from $n = 2$ to $n = 3$ is the dominant determinant of where the observed codes sit relative to their randomized backgrounds. (b) Position-specific synonymous fraction for each condition: the fraction of single-nucleotide substitutions at each codon position that preserve the coded amino acid (or RAA class). In the triplet codes (B and C), position 3 absorbs a majority of substitutions as synonymous (68.9 % for SGC, 76.5 % for RAA10 triplet), creating an error- buffering corridor. The doublet code has no dedicated synonymy position (33.3 % at pos 1, 8.3 % at pos 2; pos 3 is absent). N/A = not applicable (condition A has no third position). Raw z-scores are reported; no inter- condition z-score differences (Δz) are computed because z-scores against different null models are not directly comparable.

Triplet: GCU (Ala) neighborhood



Doublet: GC (Ala/Gly) neighborhood

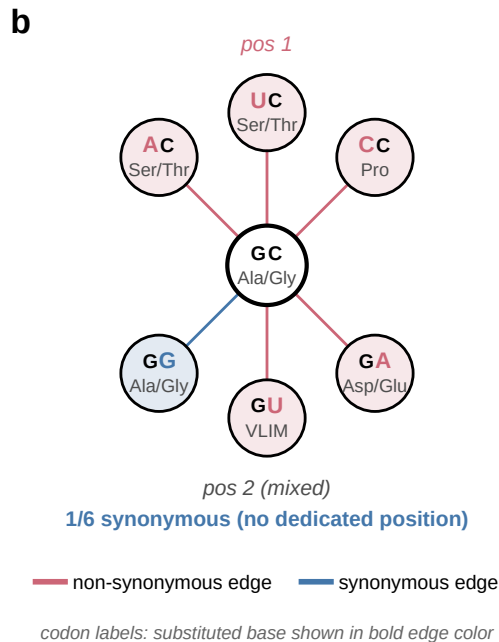


Figure 4. Wobble-position synonymy is the mechanism by which triplet architecture buffers single-nucleotide errors. (a) Hamming-1 neighborhood of GCU (Ala) in the SGC. Each neighbor differs from GCU by exactly one base (the substituted base is shown in bold in the edge color). The three position-3 neighbors (GCA, GCC, GCG) all encode Ala and are therefore synonymous with GCU (blue edges). All six position-1 and position-2 neighbors encode different amino acids (red edges). The wobble position converts all three position-3 substitutions into silent changes; positions 1 and 2 convert none, for a neighborhood synonymy rate of 3/9 (33 %). (b) Hamming-1 neighborhood of GC (Ala/Gly), the doublet-projection analogue of GCU under the RAA10 alphabet. The doublet code has six neighbors (three at each of its two positions). Only GG (which the majority-vote projection also assigns to Ala/Gly) is synonymous; the other five neighbors cross class boundaries. Synonymy rate 1/6 (17 %), with no position providing dedicated wobble tolerance. This explains the mechanism behind the architecture effect in Figure 3: the third codon position in triplet codes supplies a systematic error-absorbing channel that has no analogue in doublet codes, regardless of alphabet size.

Supplementary Materials

Companion to “Triplet architecture enables deep error-minimization of the genetic code”

Submitted to the Journal of Molecular Evolution.

Table S1 — Sensitivity of Condition C to representative-set choice

Condition C (10-class triplet code at $n = 3$) was re-evaluated using three strategies for selecting one representative amino acid per RAA10 class: closest to centroid (used in the main text), farthest from centroid, and random class member. For the random strategy, three independent seeds (1, 2, 42) were run and the reported value is the mean across seeds $\pm$ sample standard deviation. Each entry was computed against its own 100 000-shuffle null distribution.

Representative set	Distortion z (D)	Dirichlet z (E)	D-E correlation
Closest to centroid (main text)	−14.79	−13.34	0.958
Farthest from centroid	−14.61	−13.18	0.965
Random class member (mean $\pm$ SD, 3 seeds)	−15.17 $\pm$ 0.16	−14.09 $\pm$ 0.54	0.961 $\pm$ 0.008

Per-seed values for the random strategy:

- D z-scores: seed 42 = −15.09, seed 1 = −15.36, seed 2 = −15.07
- E z-scores: seed 42 = −13.74, seed 1 = −14.71, seed 2 = −13.83
- D-E correlations: seed 42 = 0.955, seed 1 = 0.969, seed 2 = 0.958

Z-score ranges across strategies (using the random-strategy mean):

- Distortion z: **0.56**
- Dirichlet z: **0.92**

Both ranges are small compared to the ~ 15 SD displacement of the SGC from its null under both metrics, so the qualitative conclusion (SGC lies far outside the null under any reasonable choice of representative set) does not depend on the representative-selection rule.

Sources: results/publication_controls.json (closest, farthest, random seed 42); results/sensitivity_extra_seeds.json (random seeds 1 and 2); aggregation via build_table_s1.py.

Table S2 — Physicochemical descriptor set used for PCA

The amino acid property library begins with 22 physicochemical descriptors and retains 20 after correlation pruning at $|r| \geq 0.95$. The retained

descriptors are then standardized (z-scored across the 20 amino acids) and compressed by PCA; the first 8 principal components explain 97.1 % of total variance and are used as the property vector $f(\cdot)$ throughout.

Retained descriptors (20):

1. `hydro_kd_z` — Kyte-Doolittle hydropathy
2. `vol_zam_z` — Zamyatnin side-chain volume
3. `polarity_gr_z` — Grantham polarity
4. `pI_z` — isoelectric point
5. `sc_pka_z` — side-chain pKa
6. `charge_pH7_z` — net side-chain charge at pH 7
7. `donors_z` — hydrogen-bond donor count
8. `acceptors_z` — hydrogen-bond acceptor count
9. `arom_z` — aromaticity indicator (0/1)
10. `ring_count_z` — number of rings
11. `rotb_z` — rotatable bond count
12. `bulkiness_z` — Zimmerman bulkiness
13. `cf_alpha_z` — Chou-Fasman α -helix propensity
14. `cf_beta_z` — Chou-Fasman β -sheet propensity
15. `cf_turn_z` — Chou-Fasman turn propensity
16. `acc_pref_z` — relative solvent accessibility
17. `flex_z` — normalized flexibility index
18. `helical_moment_z` — helical moment proxy
19. `beta_moment_z` — β -sheet moment proxy
20. `sc_mass_z` — side-chain mass

Dropped during correlation pruning (2 descriptors):

- refractivity ($|r| \geq 0.95$ with bulkiness)
- `hphob_fr` ($|r| \geq 0.95$ with `hydro_kd`)

PCA summary:

- Retained descriptors: 20
- Principal components retained: 8
- Cumulative variance explained: 0.9710

Sources: `src/io/aa_props_lib.py` (descriptor definitions); `build_aa_props.py` (pruning + PCA pipeline); `data/processed/aa_props_meta.json` (variance breakdown).

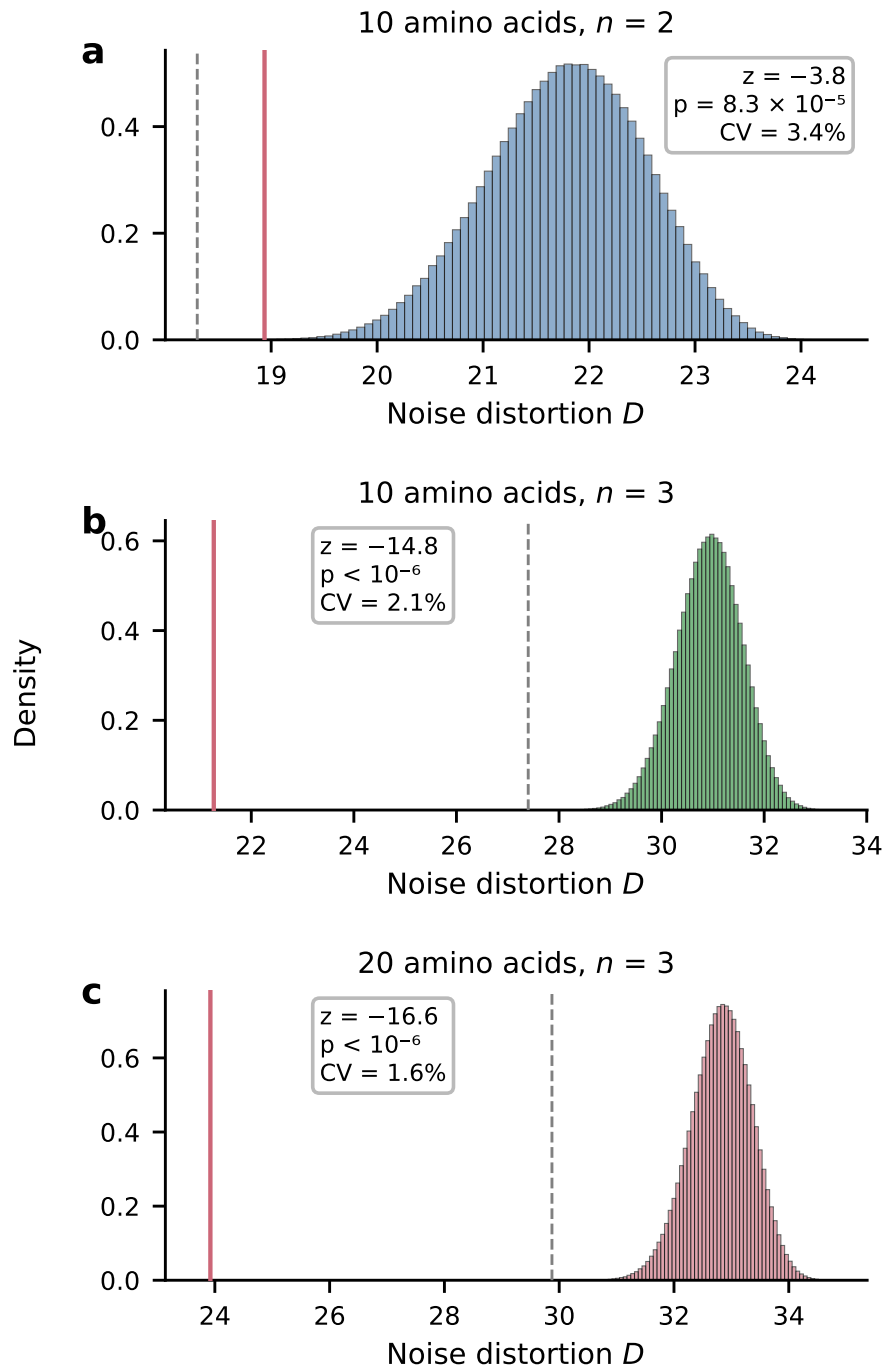


Figure S1. Raw null distributions of noise distortion D for the three conditions used in the factorial analysis. Each null was generated by 10^6 degeneracy-preserving shuffles of amino-acid (or RAA10 class) labels across the sense codons, preserving per-class codon counts, fixing stops, and pinning AUG $\rightarrow$ Met. (A) Condition A: 10-class doublet ($n = 2$) code obtained by majority-vote projection of the SGC under the RAA10 alphabet; the SGC projection's observed value (solid red line) lies within the left tail of the null ($z = -3.8$, $p = 8.3 \times 10^{-5}$), with coefficient of variation $CV = 3.4\%$. (B) Condition C: 10-class triplet ($n = 3$) code with RAA10 applied to the sense codons; the observed value (solid red line) lies outside the null ($z = -14.8$, $p < 10^{-6}$, $CV = 2.1\%$). (C) Condition B: the 20-AA SGC at $n = 3$; the observed value lies outside the null ($z = -16.6$, $p < 10^{-6}$, $CV = 1.6\%$). In all three panels the dashed grey line marks the minimum value observed across the 10^6 null samples; the solid red line marks the SGC's observed value. In (B) and (C) the red line sits well to the left of the dashed grey line — no random code in 10^6 samples reaches the SGC's noise distortion. All CV values are verified against `results/publication_controls.json``: $A = 0.0343$, $C = 0.0211$, $B = 0.0164$ (3.4 %, 2.1 %, 1.6 % when rounded to one decimal).

File manifest (supplementary files submitted alongside the manuscript)

Tables (raw data and typeset versions):

- `supplementary/table_S1.csv` — Table S1 raw values (closest, farthest, random-seeds mean $\pm$ SD)
- `supplementary/table_S1.tex` — Table S1 in LaTeX booktabs format
- `supplementary/table_S1_summary.json` — machine-readable numeric summary including per-seed values
- `supplementary/table_S2.csv` — Table S2 descriptor set as raw values
- `supplementary/table_S2.tex` — Table S2 in LaTeX booktabs format

Figures:

- `figures/figS1_raw_distributions.png` — Figure S1 at 300 DPI
- `figures/figS1_raw_distributions.pdf` — Figure S1 vector version with editable text (fonttype 42)

Supporting data (optional; can be provided on request or via a repository):

- `results/publication_controls.json` — full factorial + sensitivity numerical output (closest, farthest, random seed 42)
- `results/sensitivity_extra_seeds.json` — random seeds 1 and 2 used for the 3-seed mean in Table S1
- `results/null_distributions.npz` — raw 10^6 null distributions for all conditions (~46 MB, not typically included with submission; regeneratable via `python run_publication_controls.py --n-null 1000000 --workers 32`)

Scripts (reproducibility, in the companion code repository):

- `run_publication_controls.py` — runs the 2×2 factorial + sensitivity
- `run_sensitivity_extra_seeds.py` — runs the additional random seeds for Table S1
- `build_table_s1.py` — aggregates the three random seeds into Table S1